

KATUNAYAKE, Sri Lanka

WMO: 434500

Lat: 7.167N

Lon: 79.883E

Elev: 8

StdP: 101.23

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.0	22.0	15.9	11.3	29.4	17.1	12.2	27.7	9.2	29.1	8.4	28.9	2.9	40	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	4.9	33.1	24.7	32.5	25.1	32.1	25.3	27.6	30.8	27.2	30.6	26.9	30.3	5.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	30.1	26.2	21.6	29.6	26.0	21.4	29.5	87.9	31.0	86.3	31.0	84.9	30.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	6.8	DB	18.9	35.0	1.0	1.0	18.2	35.7	17.6	36.3	17.1	36.9	16.4	37.6	(4)
(5)			WB	17.4	27.9	0.9	0.6	16.7	28.3	16.2	28.7	15.6	29.0	14.9	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	26.9	27.5	28.1	28.3	28.6	28.2	28.0	27.9	27.8	27.3	27.0	26.8	(6)	
	DBStd	1.10	1.07	1.14	0.88	1.01	1.04	0.94	0.85	0.87	0.90	0.85	0.81	0.89	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6462	523	489	561	551	578	546	558	555	533	537	511	522	(10)	
	CDD18.3	3421	265	256	302	301	319	296	299	297	283	279	261	263	(11)	
	CDH23.3	36668	2458	2623	3483	3464	3923	3481	3343	3256	3098	2838	2367	2334	(12)	
	CDH26.7	12829	889	1033	1452	1462	1603	1199	987	939	937	860	724	743	(13)	
(14) Wind	WSAvg	3.6	4.1	3.4	2.8	2.7	3.8	4.3	4.3	4.3	4.0	3.0	2.6	3.5	(14)	
(15) Precipitation	PrecAvg	2031	32	61	133	237	320	139	112	69	157	367	276	128	(15)	
	PrecMax	3498	146	193	268	497	682	322	458	280	353	700	537	509	(16)	
	PrecMin	1125	0	0	13	54	26	14	1	1	11	74	12	4	(17)	
	PrecStd	545	41	51	74	114	183	86	108	70	103	166	140	114	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	34.8	34.1	33.2	32.6	31.4	31.2	31.2	31.2	31.2	32.2	32.8	(19)	
		MCWB	22.9	23.4	24.4	26.9	27.1	27.0	26.5	26.0	25.9	25.7	24.2	23.5	(20)	
	2%	DB	32.2	33.2	32.9	32.8	32.1	31.1	30.2	30.7	30.9	30.9	31.2	31.8	(21)	
		MCWB	23.0	23.2	25.4	26.7	27.0	26.8	26.0	25.9	25.9	25.7	24.8	23.7	(22)	
	5%	DB	31.8	32.1	32.1	32.1	31.8	30.4	30.1	30.1	30.2	30.2	30.8	31.0	(23)	
		MCWB	23.1	23.8	25.5	26.5	26.9	26.4	25.9	25.8	25.7	25.6	25.0	24.0	(24)	
	10%	DB	30.9	31.2	31.3	31.8	31.1	30.1	29.9	29.9	30.0	30.0	30.0	30.1	(25)	
		MCWB	23.3	24.3	25.5	26.4	26.8	26.2	25.8	25.7	25.7	25.6	25.1	24.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.1	26.6	27.2	27.9	28.2	27.8	27.0	26.8	26.8	26.7	26.7	26.3	(27)	
		MCDB	29.7	30.7	31.5	31.8	31.0	30.3	30.0	29.6	29.8	29.8	30.0	29.5	(28)	
	2%	WB	25.5	26.1	26.7	27.4	27.7	27.1	26.5	26.4	26.4	26.4	26.2	25.8	(29)	
		MCDB	29.2	29.9	30.9	31.4	30.7	29.9	29.4	29.2	29.5	29.4	29.5	29.2	(30)	
	5%	WB	25.0	25.7	26.4	27.0	27.3	26.7	26.2	26.1	26.1	26.1	25.8	25.4	(31)	
		MCDB	28.9	29.6	30.5	31.0	30.5	29.5	29.0	29.0	29.2	29.1	29.1	28.8	(32)	
	10%	WB	24.6	25.2	26.1	26.7	27.0	26.5	26.0	25.9	25.9	25.8	25.5	25.0	(33)	
		MCDB	28.4	29.3	30.2	30.6	30.3	29.3	28.8	28.7	29.0	28.8	28.7	28.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	8.6	7.6	6.7	4.9	3.9	3.6	3.9	4.5	5.4	6.5	7.2	(35)	
		MCDBR	9.8	9.9	8.2	7.2	5.4	4.2	3.9	4.2	4.8	5.8	7.6	8.4	(36)	
	5% WB	MCWBR	3.2	3.5	3.3	3.0	2.4	2.1	1.9	1.8	2.1	2.5	3.0	3.0	(37)	
		MCWBR	7.7	8.0	7.4	6.7	4.9	4.2	3.9	4.1	4.6	5.4	6.5	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.432	0.444	0.489	0.501	0.522	0.500	0.515	0.497	0.466	0.474	0.465	0.452	(40)	
		taud	2.374	2.322	2.212	2.205	2.134	2.209	2.142	2.212	2.314	2.292	2.316	2.336	(41)	
	Ebn at Noon		884	882	843	820	785	791	783	812	850	844	846	856	(42)	
		Edh at Noon	124	134	150	149	156	142	153	146	134	136	130	127	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.25	5.96	6.33	6.00	5.42	5.36	5.49	5.60	5.66	5.36	4.69	4.66	(44)	
	RadStd		0.39	0.32	0.28	0.29	0.40	0.29	0.27	0.27	0.36	0.31	0.35	0.41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.17	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page